

Deep Learning



Lecture 01: Introduction

Overview



Today:



Administrative Information

Instructors Information
Course Information
Textbooks
Grading Criteria



Next Week:

Deep Learning Introduction

Course Instructor

- Dr. Shahbaz Khan
- Assistant Professor
- Department of Robotics and Artificial Intelligence, SMME, NUST.
- Email: Shahbaz.khan@smme.nust.edu.pk
- Office: 215 C
- Office Hours: Tuesday & Thursday, 12:00 - 4:00 pm or by appointment



Let's get to know each other....



MYSELF !



YOU !



THE SUBJECT !

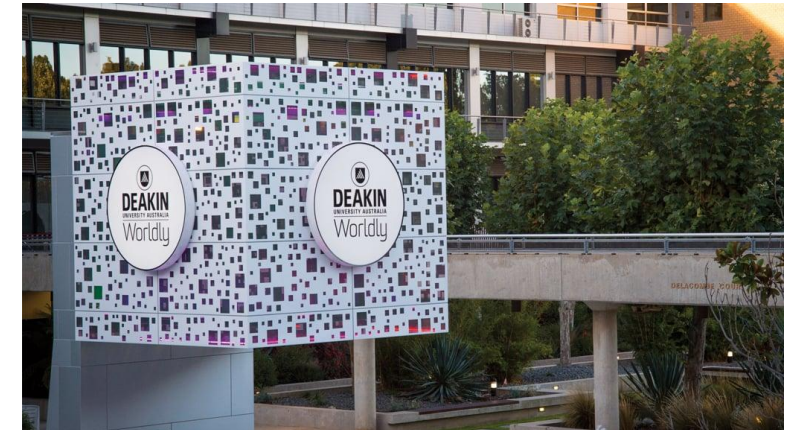
- **Qualification**

- **BE. Mechatronics Engineering**
 - UET Peshawar
- 2008-2012



- **Masters (Mechanical Engineering & Engineering Management)**
 - Deakin University Geelong Waurin Pond Campus Australia
- 2014-2016

- **PhD Robotics and AI**
 - UET Peshawar
- 2019-2023
- **Research Area**
 - Robotics (UAV) and AI in Precision Agriculture



- **Professional Experience**

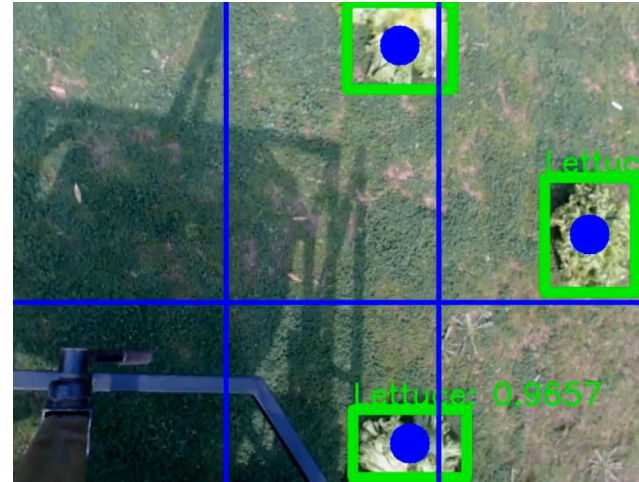
- Lecturer at UEAS (Feb 2022-Jan 2023)
- Lab Engineer at UET Peshawar (2019-Jan 2022)
- Lecturer Hitec University Taxila (2017-2019)
- Consultant for Stemit.
- Consultant of different private automation industries.
- Member of ARAL LAB (NCRA)
- Consultant for STEM LAB UET Peshawar.

Golden Key International Society member from Deakin University Chapter in 2016.

Who Am I?

- **Research Profile:**
- **Publications:**
 - 18 Journals
 - 3 Book Chapters
- **Reviewer of Journals:**
 - Frontier in Plant Sciences.
 - Agriculture MDPI
 - Drones MDPI
 - IEEE Access

Research Work in ARAL Lab





Robotics Competitions

Founder of
HRT, YRT &
KRT



Subjects I have thought at the UG & PG Level

Measurement
and
Instrumentation

Robotics

Industrial
Robotics

Applied Physics

Control
Engineering

Microprocessors
and Micro
Controllers

Introduction to
Mechatronics

Research
Methodology

Mechatronics
System Design

Engineering
Management

Project
Management

Innovation &
Technology
Management

Deep Learning

AI for
Accountants



WHAT IS YOUR
NAME?



WHERE ARE YOU
FROM?



WHAT IS YOUR GPA
AND YOUR
BACKGROUND



YOUR HOBBIES



GOOD AND BAD
THING ABOUT
UNIVERSITY LIFE



YOUR EXPECTATIONS
FROM THE SUBJECT?



FUTURE GOALS

**What
About
You?**

This course is not easy

- A lot of work!
- A lot of work!!

This course is not easy

- A lot of work!
- A lot of work!!
- A lot of work!!!

This course is not easy

- A lot of work!
- A lot of work!!
- A lot of work!!!
- **A LOT OF WORK!!!!**

Not for chickens!



This course is not easy

- A lot of work!
 - A lot of work!!
 - A lot of work!!!
 - **A LOT OF WORK!!!!**
-
- But a lot of fun :)



This course is not easy

- *Mastery-based* evaluation
 - **Quizzes & Exams** to test your understanding of topics covered in the lectures
 - Weekly Tutorials to teach you to implement algorithms
 - **Assignments** are designed to guide you through an extensive review of existing literature, enabling you to critically analyze prior work and systematically identify potential research gaps in the chosen topic
 - **Project** is designed to help you take a research paper and bring it to life by implementing its concepts step by step and expose you to real world DL problems
 - **FYI: Anyone who gets an “A” in the course is technically ready for a deep learning research or job**



وَيَسْأَلُونَكَ عَنِ الرُّوحِ قُلِ الرُّوحُ مِنْ أَمْرِ رَبِّي وَمَا أُوتِيتُمْ مِنَ الْعِلْمِ إِلَّا قَلِيلًا ﴿٨٥﴾

They ask you 'O Prophet' about the spirit. Say, "Its nature is known only to my Lord, and you 'O humanity' have been given but little knowledge."

• <i>Quizzes</i>	<i>10%</i>
• <i>Assignments</i>	<i>10%</i>
• <i>Project</i>	<i>15%</i>
• <i>Mid Term</i>	<i>25%</i>
• <u><i>Final Exam</i></u>	<u><i>40%</i></u>
<i>Total</i>	<i>100%</i>

Note: Strict Action will be taken for copying the assignments.
The grading scheme is *tentative*. The instructor reserves the right to change it.

- In-class lectures
 - Lectures will be uploaded on MS teams
 - First half would be lecture
 - After Lecture, a step-by-step tutorial needs to be implemented in the class itself.
 - The tutorial will be evaluated in the next class, and the marks will be added in the **5%** of Project at the end of the course.
 - Two Students will **Present** their progress weekly in the last **20 minutes** of the class.
- Importance of viewing lectures
 - Even if you think you know the topic
 - Quizzes: your marks depend on viewing lectures
 - Monitor attendance (more on this later)

Assignment 1 (Deadline 5th Week)

- Review **10** research papers
- Focus on: Methodology, Limitations, Results

Assignment 2 (Deadline 10th Week)

- Review **20** research papers
- Identify common themes, gaps, and future directions

Assignment 3 (Deadline 15th Week)

- Review **20** more research papers
- Deeper focus on comparative analysis of methods and outcomes
- Final Submission **Write a complete review paper**
- Highlight the research gap, proposed future work, and a cohesive synthesis of all reviewed literature

Project 1 - Journal Paper Implementation (16th Week) (10%)

- Reproduce results of a journal paper (published between 2023 till date)
- Demonstrate understanding of the methodology and experiments

Project 2 - Tutorial-Based Development (After Mid Exam) (2.5%)

- Select a project idea from **course tutorials**
- Extend it into a **working project**

Weekly Tutorials (After Every Class) (2.5%)

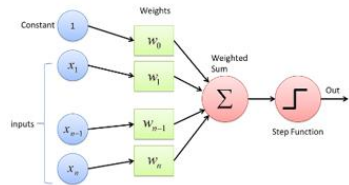
- Completion of weekly tutorials will contribute **2.5%** each toward the **final project grade**
- Ensures steady progress and application of concepts

Tutorial 1: Building a Perceptron

Objectives:

- Building Perceptron from Scratch
- How Weights and Bias get Updated
- Activation Function
- Data Loading
- Training & Testing

Lets build a Perceptron?



Step 1 – Perceptron Class

Follow these simple steps:

- Create a file called “Perceptron.py”
- Define a class called Perceptron inside

```
import numpy as np

class Perceptron(object):
    def __init__(self, eta=0.01, n_iter=10):
        self.eta = eta
        self.n_iter = n_iter
```

To build a perceptron, we need 3 attributes:

- η (eta)**: the learning rate is usually a small value between 0.0 and 1.0 which defines how quickly the model learns.
- n_iter**: the number of iterations.
- w_**: the weights. A weight defines the importance of the corresponding input value

Step 2 – Weighted sum

```
def weighted_sum(self, X):
    return np.dot(X, self.w_[1:]) + self.w_[0]

# np.dot(X, self.w_[1:]) computes the dot product of the input features 'X' and the model's weights (excluding the bias weight)
# self.w_[0] represents the bias term (also known as the intercept), which is added to the dot product
```

np.dot(X, w_) is simply:

$$\sum_{i=0}^n X_i \cdot w_i$$

Step 3 – Step function

```
def predict(self, X):
    return np.where(self.weighted_sum(X) >= 0.0, 1, -1)
```

The step function predict () returns:

- 1 – if the weighted sum is greater than 0.0
- 1 – Otherwise.

Step 4– Create the training method

An explanation will follow.

```
def fit(self, X, y):
    # Initializing the weights to 0
    self.w_ = np.zeros(1 + X.shape[1]) # Weights initialized to zeros, with one more element for the bias
    self.errors_ = [] # List to track the number of errors in each iteration
    print('Weights:', self.w_) # Print the initial weights for reference

    # Training the model for 'n_iter' number of iterations
    for i in range(self.n_iter):
        # Loop for the total number of iterations
        error = 0 # Initialize error counter for the current iteration

        # Loop through each input and corresponding target value
        for xi, target in zip(X, y):
            # Predict the output using the current weights
            # Predict the output for the current input 'xi'
            pred = self.predict(xi)

            # Calculate the weight update
            # Update 'w_' is the learning rate
            update = self.eta * (target - pred) # Update the update value based on prediction error

            # Update the weights for each input
            # Update 'w_' is the learning rate
            self.w_[1:] += update * xi # Update the weights (excluding bias)
            print('Updated Weights:', self.w_[1:]) # Print updated weights for debugging

        # Update the bias (w_0)
        self.w_[0] += update # Update the bias term separately

        # Calculate the error count of the model's predictions and update
        error += np.sum(np.abs(target - pred)) # If update is 0, it means the prediction was wrong

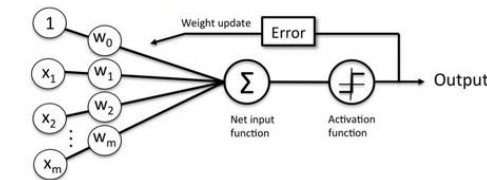
    # Return the total number of errors in this iteration to the array 'self.errors_'
    self.errors_.append(error) # Store the error count after each iteration
```

We initialize the weights to 0, then start our training.

For each input:

- we calculate $\hat{y}$, the predicted output.
- then we update the weights from w_1 to w_0 (remember w_0 is the bias).

We repeat this process n_iter times (n_iter is the number of iterations).



Weights update in the perceptron

Step 8: Measuring Performances

```
from sklearn.metrics import accuracy_score

# Importing the accuracy_score function from sklearn's metrics module.
# This function is used to calculate the accuracy of the model's predictions.

y_preds = perceptron.predict(test_data)

# Using the trained perceptron model to make predictions on the test data.
# The predict() method returns an array of predicted labels for the test dataset (test_data).
# The predicted labels are stored in y_preds.

# Calculate the accuracy of the model by comparing predicted labels (y_preds) with actual test labels (test_labels)
accuracy = accuracy_score(y_preds, test_labels)

# The accuracy_score function compares the predicted labels (y_preds) with the actual labels (test_labels).
# It returns the proportion of correct predictions, which is the accuracy of the model.

# Print the accuracy, rounded to two decimal places, and multiplied by 100 to show it as a percentage.
print('Accuracy:', round(accuracy, 2) * 100, '%')

# The accuracy is rounded to 2 decimal places, then multiplied by 100 to convert it to percentage format for readability.
```

Tasks:

- Take manual input for the four features (sepal length, sepal width, petal length, and petal width).
 - Use the model to predict whether it's Iris-setosa or not based on the manually provided values.
 - Display the prediction result.
- Replace the Step Function with Sigmoid.
- Replace the labels of 1,-1 with 1 & 0.

Required textbooks

- Deep Learning by Ian Goodfellow and Yoshua Bengio and Aaron Courville
(<https://www.deeplearningbook.org/>)
- Hands-On Machine Learning with Scikit-Learn and TensorFlow Concepts, Tools, and Techniques to Build Intelligent Systems by Aurélien Géron, 1st Edition, March 2017.
- Deep learning with python by Francois Chollet

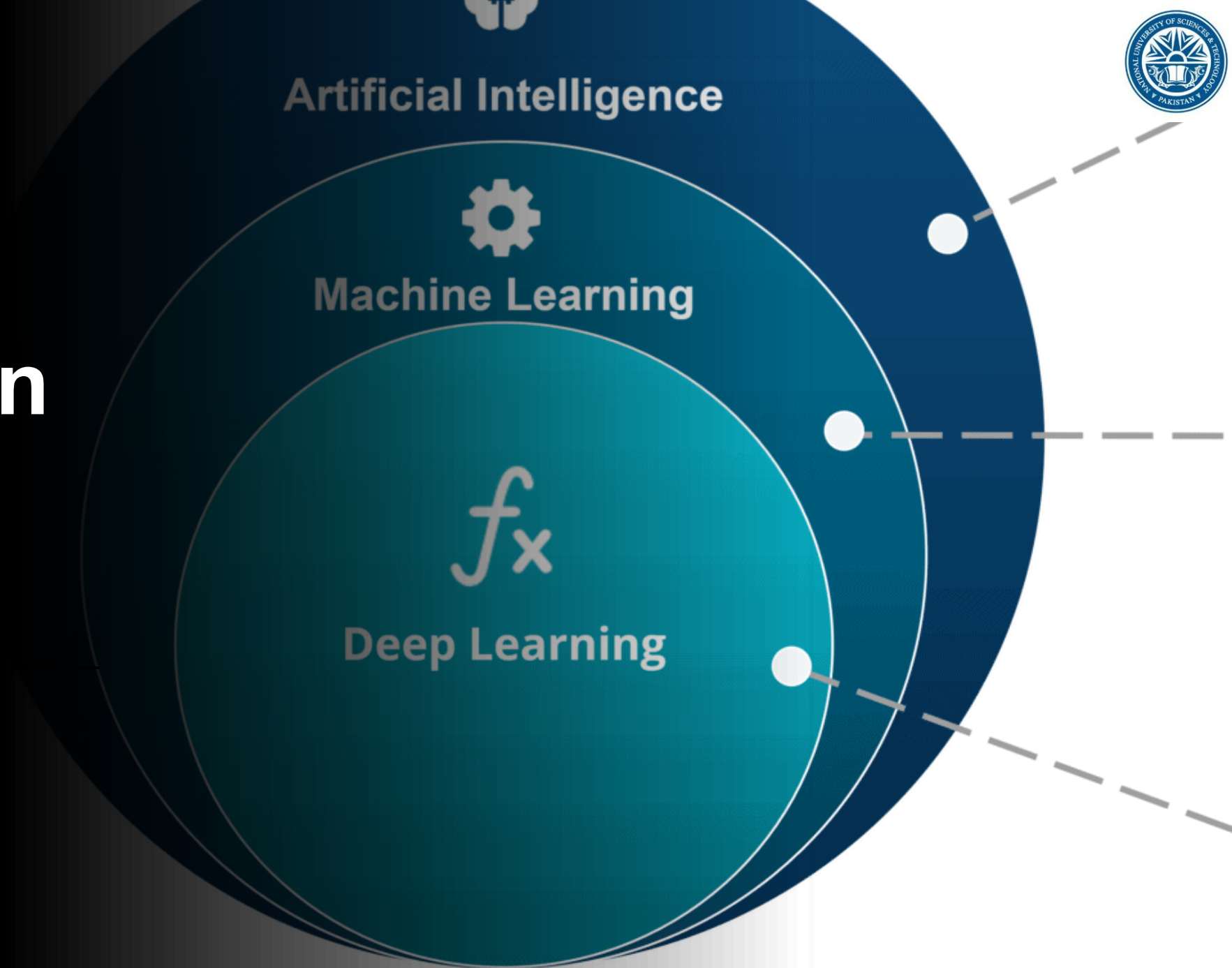
Tools

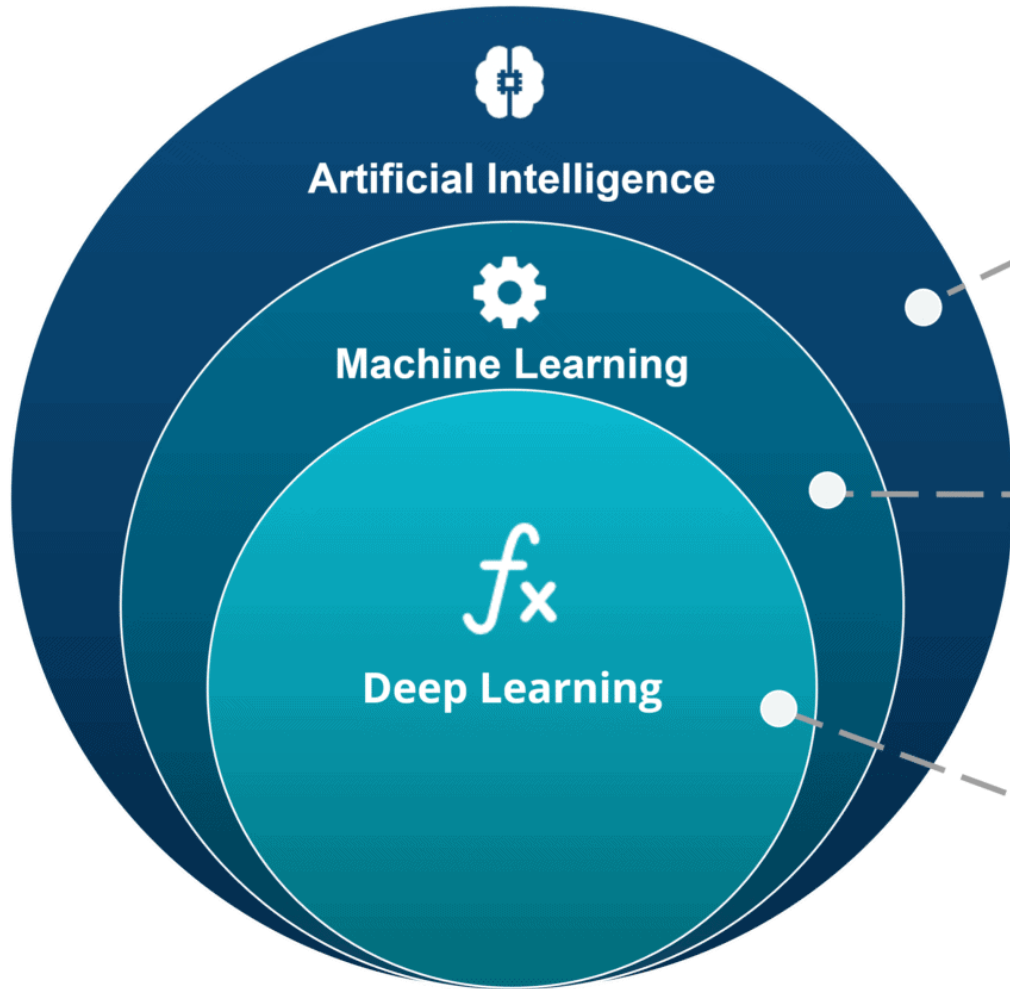


Classroom Etiquettes :

- It is the collective responsibility of all the students to make the class environment conducive to learning. To create and maintain a friendly atmosphere, the following standards of classroom behavior will be observed,
 - Students will be punctual for the class. The teacher considers latecomers disrespectful of those who manage to be on time.
 - If a student decides to attend the class, he or she will not disrupt class by leaving before the lecture has ended.
 - All the cell phones must be switched OFF before entering the classroom.

Introduction to Deep Learning





ARTIFICIAL INTELLIGENCE

A technique which enables machines to mimic human behaviour

MACHINE LEARNING

Subset of AI technique which use statistical methods to enable machines to improve with experience

DEEP LEARNING

Subset of ML which make the computation of multi-layer neural network feasible

Timeline

1950's	 Artificial intelligence (AI) <i>Human intelligence exhibited by machines</i>		
	1980's	 Machine learning <i>AI systems that learn from historical data</i>	
		2010's	 Deep learning <i>Machine learning models that mimic human brain function</i>
		2020's	 Generative AI (Gen AI) <i>Deep learning models (foundation models) that create original content</i>

Breakthroughs

www.technewsworld.com/story/84013.html

40 maps that explain Amazon Web Services Primers | Math n Procs deeplearning.net/tuto Deep Learning Tutori deep learning PHILIPS - Golden Ears Language Technology MyIDCare - Dashbo Other bookmarks

TECHNEWSWORLD EMERGING TECH

Computing Internet IT Mobile Tech Reviews Security Technology Tech Blog Reader Services

Microsoft AI Beats Humans at Speech Recognition

By Richard Adhikari
Oct 20, 2016 11:40 AM PT

Print
Email

5
25
45
11
0
104



Image: Adobe Stock

Microsoft's Artificial Intelligence and Research Unit earlier this week reported that its speech recognition technology had surpassed the performance of human transcriptionists.

How do you feel about Black Friday and Cyber Monday?

- They're great -- I get a lot of bargains!
- The deals are too spread out -- I'd prefer just one day.
- They're a fun way to kick off the holiday season.
- I don't like the commercialization of Thanksgiving Day.
- They're crucial for the retail industry and the economy.
- The deals typically aren't that good.

Vote to See Results

E-Commerce Times

Black Friday Shoppers Hungry for New Experiences, New Tech

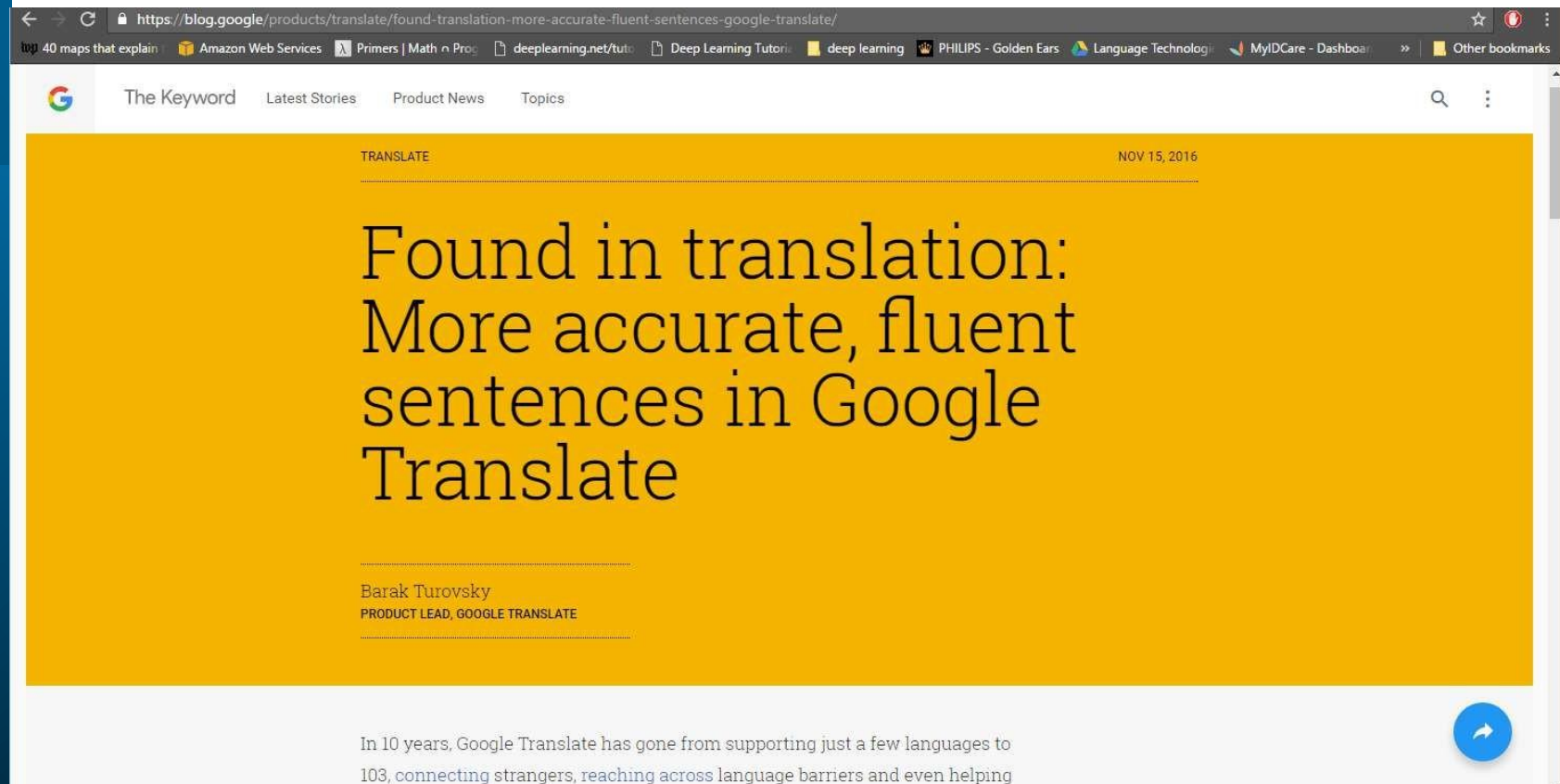
Pay TV's Newest Innovation: Giving Users Control

Apple Celebrates Itself in \$300 Coffee Table Tome

AWS Enjoys Top Perch in IaaS, PaaS Markets

US Comptroller Gears Up for Blockchain and

Breakthroughs



And Now

ChatGPT, tell me about this picture

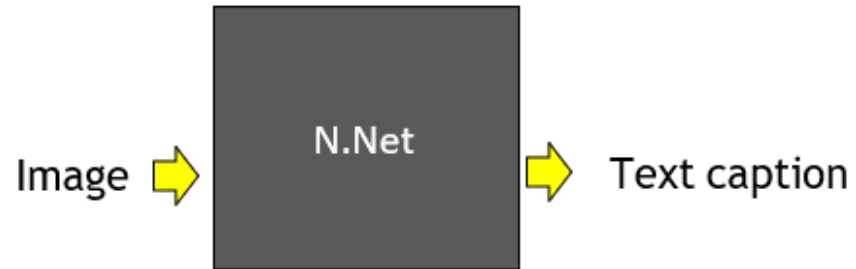
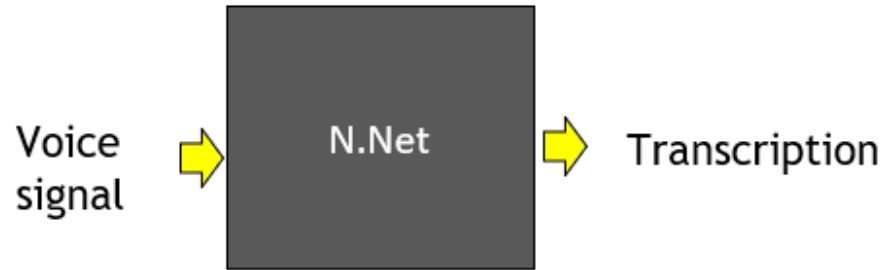


7:12 PM The ceiling in the photo appears to have several noticeable water stains, which could indicate potential water damage. The stains seem to be spread across multiple tiles, suggesting there might be a leak from the roof or pipes above. It would be advisable to address this issue promptly to prevent any further damage or potential mold growth. Notifying building maintenance or the landlord to investigate and fix the source of the leak would be the best course of action.

Besides the water stains on the ceiling, the photo shows a few other details:

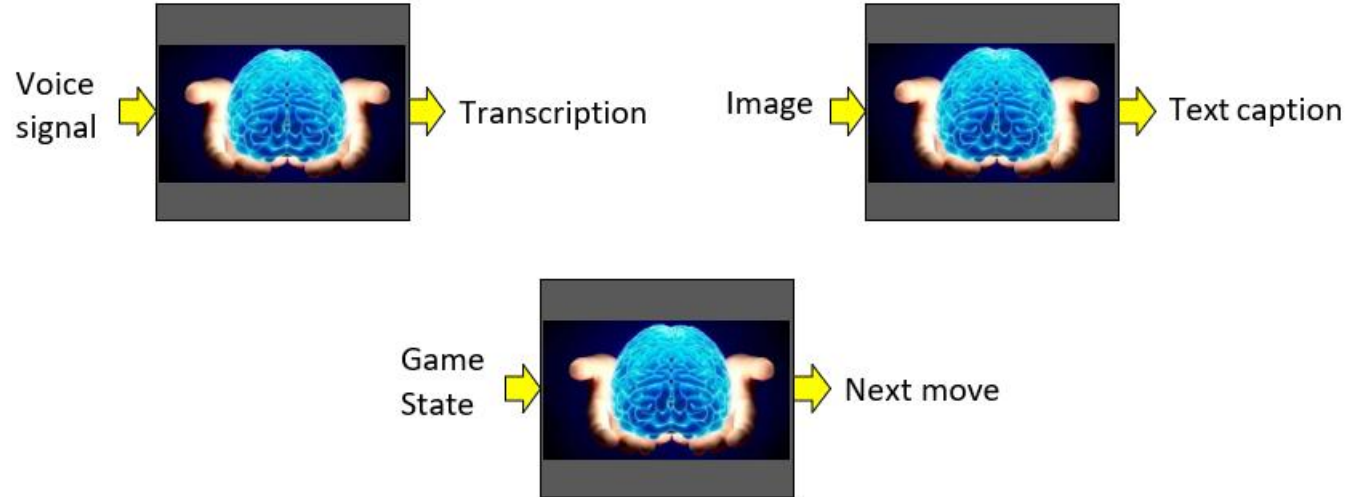
1. **Posters on the Wall:** There are several posters or prints on the wall featuring various images, likely of scenes from movies, video games, or other forms of media.
2. **Furniture and Clutter:** The room contains a few pieces of furniture, including a blue chair, a black wardrobe or storage unit, and a desk with two monitors. There are clothes and other items visible on the wardrobe and desk, indicating a typical lived-in space.
3. **Lighting and Ventilation:** The ceiling has a light fixture and an air vent, suggesting this is a standard room with typical building amenities.
4. **Bed and Bedding:** The lower part of the photo shows a bed with a person lying on it, with their foot visible in the foreground. The bed has a comforter and some clothing or linens on it.
5. **Overall Condition:** The room appears somewhat cluttered, with personal belongings spread out. The presence of water stains on the ceiling is the most concerning issue visible.

So, What are They?



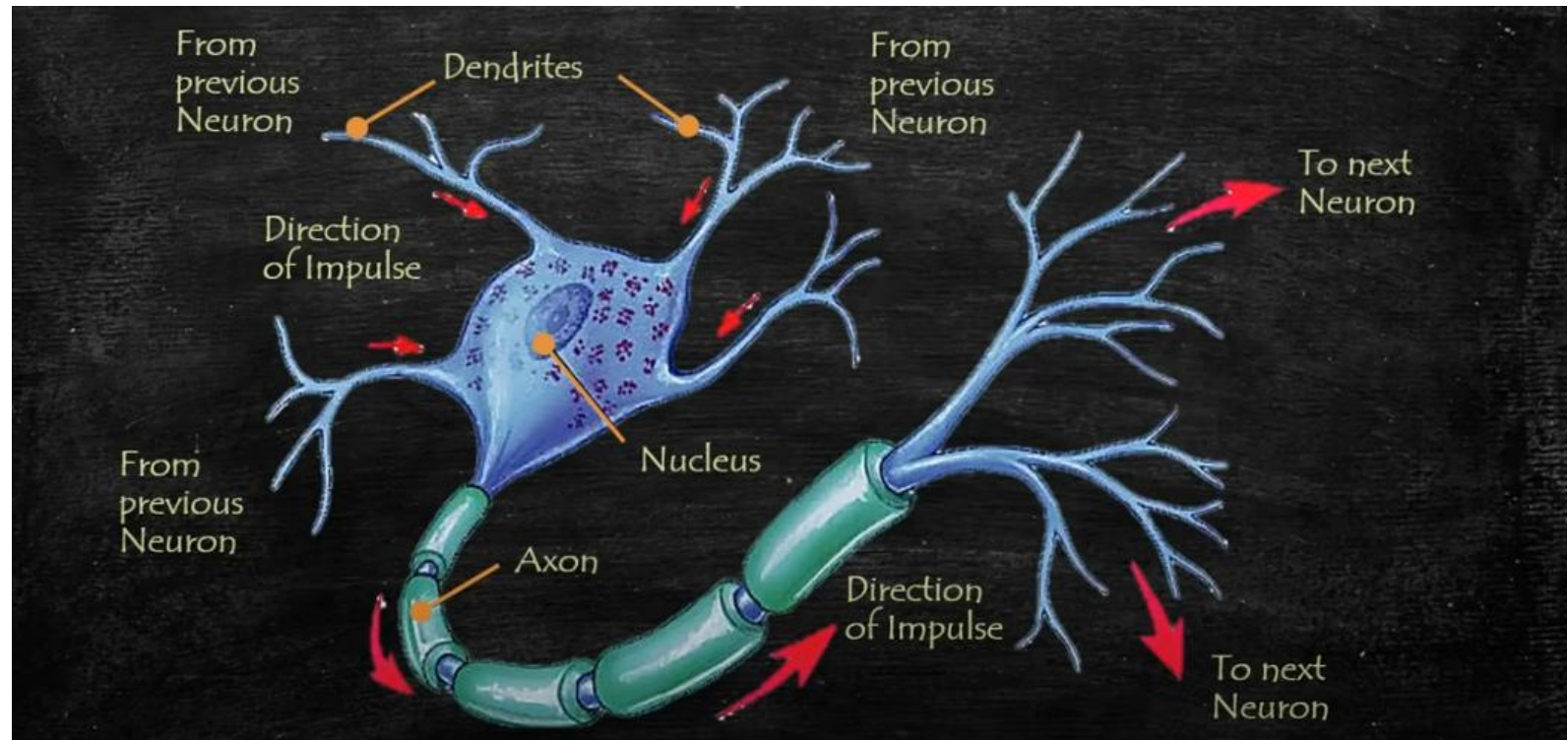
- What are these boxes?
 - Functions that take an input and produce an output
 - What are these functions?

So, What are They?



- In a human, those functions are computed by the brain...
- In their basic form, NNets mimic the networked structure in the brain

So, What are
They?



Deep Learning in Computer Vision



Sports Video
Classification



Deep Learning in Autonomous Vehicles & Robotics

Deep Sensorimotor Learning

rll.berkeley.edu/deeplearningrobotics

Department of Electrical Engineering and Computer Sciences
University of California, Berkeley

Thank You

Any Questions?